

Onkaar Grewal

Onkaar.Grewal@outlook.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Cambridge, ON

TECHNICAL SKILLS

Design & CAD: SolidWorks, Autodesk Inventor, AutoCAD, Fusion 360, KiCad, Blender

Programming & Tools: Python, C/C++, MATLAB, Arduino, LaTeX, Granta EduPack

Hardware & Fabrication: Embedded Systems, 3D Printing, Soldering, Raspberry Pi, Power Tools

PROJECTS

Portable Weather Station Node | *Arduino, C/C++, Sensors, SolidWorks* 2024 – Present

- Engineered an autonomous weather monitoring system with a sensor suite measuring **humidity, temperature, wind speed, and rainfall** with **real-time data logging**.
- Designed and fabricated a **mechanical housing in SolidWorks** to protect electronics from environmental exposure while maintaining accurate sensor readings.
- Implemented **sensor communication protocols** and **data logging pipelines** on an **Arduino-based embedded system**.

Wheelchair-Mounted Assistive Launcher | *SolidWorks, Granta EduPack, Python* 2025

- Led **mechanical CAD design** of a launcher achieving a **6m+ range**, utilizing **Granta EduPack** for structural **material selection** to benefit a client with arthritis.
- Coordinated **technical design integration** across a **multidisciplinary team** within McMaster's Engineering Design course (ENGINEER 1P13).

PID Controller Simulation | *Python, NumPy, Matplotlib* 2025

- Developed a **PID controller simulation** in Python modeling a **second-order dynamic system**, implementing **proportional, integral, and derivative tuning** to analyze transient response.
- Visualized **system response curves** using Matplotlib to compare **controller performance** across varying gain parameters.

EXPERIENCE

Engineering Design & Drafting Intern Cambridge, ON
Atlantic Industries Limited January 2023 – June 2023

- Produced **high-accuracy 2D AutoCAD drawings** for **structural civil engineering projects**, transitioning from administrative support to technical drafting.
- Collaborated with university co-op students on **large-scale infrastructure designs** and assisted in **document management** and drawing revisions.

First Robotics Competition Mentor Cambridge, ON
St. Benedict Catholic Secondary School September 2023 – January 2025

- Mentored **mechanical sub-teams** in **CAD prototyping and fabrication** of competitive **120lb robots**, guiding **drivetrain and intake design** layout.
- Instructed junior students on **shop safety** and safe operation of **power tools and industrial machinery**.

Administrative Project Support Ontario, Canada
YMCA of Three Rivers March 2025 – Present

- Coordinate **board-level initiatives and events** across **6 municipal locations**, managing **project documentation** and scheduling for regional leadership.
- Facilitate **cross-location communication** for the **Board of Directors** to ensure efficient execution of community-focused projects.

EDUCATION

McMaster University Hamilton, ON
Materials Science and Engineering (Co-op) - Year 2 Sep 2025 – Present